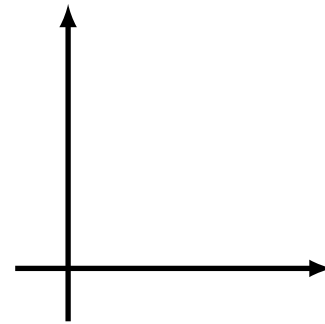
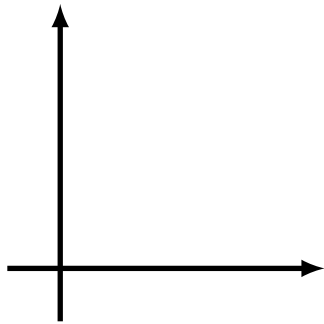


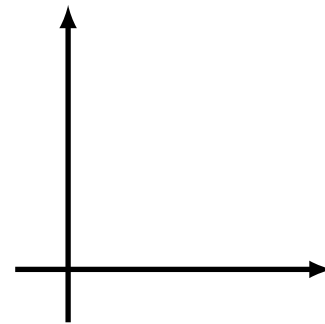
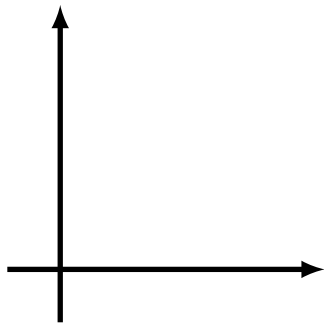
Example.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$



Example.

$$A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$$



Theorem

If A is a 2×2 matrix then the linear transformation $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ preserves orientation if $\det A > 0$ and reverses orientation if $\det A < 0$.